

Physical Pharmacy

Code: (CPT 203, PT 203)

Lecture 5

SURFACE AND INTERFACIAL PHENOMENON

Lecture Learning Outcomes (LOs)

Program Outcomes (NARS 2017)		Course Learning Outcomes	
Code	Text	Code	Text
1-1-1	Demonstrate understanding of knowledge of pharmaceutical, biomedical, social, behavioral, administrative, and clinical sciences.	a1	Recognize different physicochemical properties affecting the behavior of different types of materials (solid, liquid and gas).
1.1.3	Integrate knowledge from fundamental sciences to handle, identify, extract, design, prepare, analyze, and assure quality of synthetic/natural pharmaceutical materials/products.	a2	Classify different factors concerning physicochemical properties such as solubility , viscosity, pH, isotonicity, dissolution, partition coefficient, surface tension and others for optimizing pharmaceutical formulations
4.2.2	Use contemporary technologies and media to demonstrate effective presentation skills.	d1	Provide efficient presentation using modern technology and media

Interface:

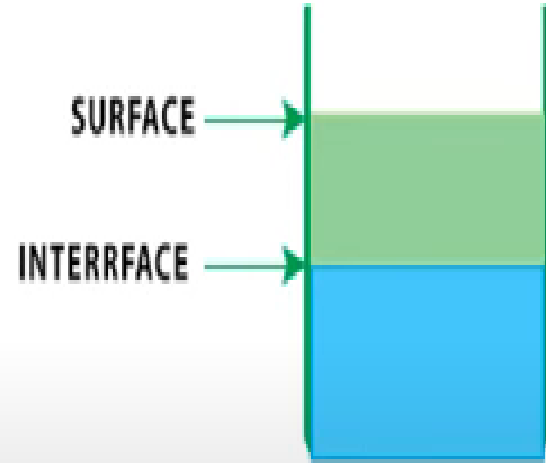
It is the boundary between two phases which exist together

Surface:

The outer most layer of any substance

It is the boundary between two phases, one of them **is air**

(Every surface is an interface)



1 & 2 are two different phases



Types and examples of interface

Several types of interface can exist depending on whether the two adjacent phases are in solid, liquid or gaseous state.

Interfaces divided into groups:

Liquid interfaces including:

1. Liquid/ Liquid (L/L) (liquid surface); e.g. oil and water (emulsion)
2. Liquid/ gas (L/V); e.g. water exposed to air

Solid interfaces including:

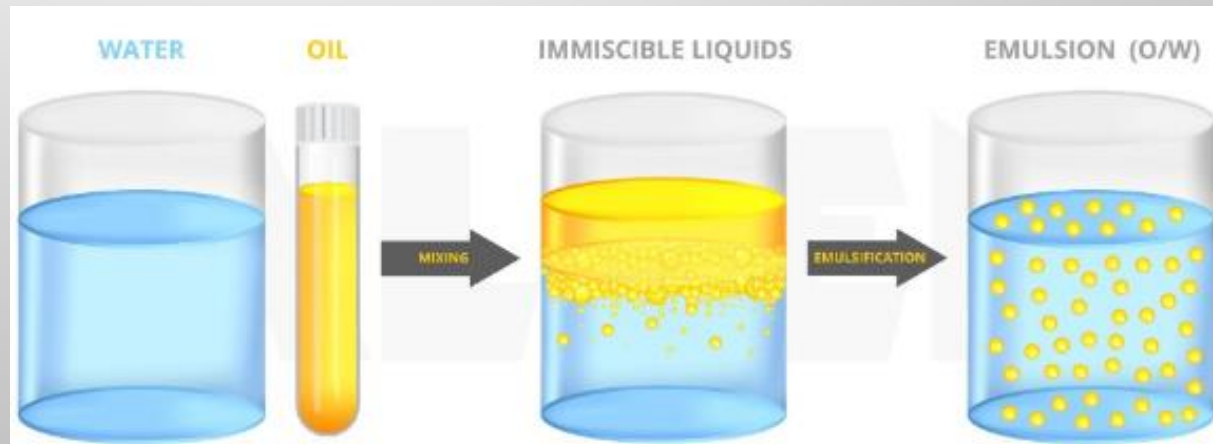
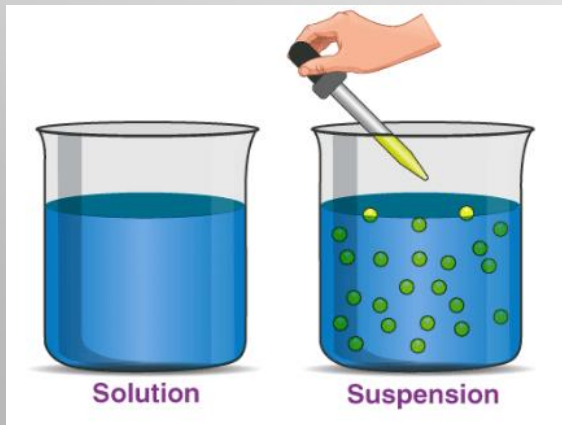
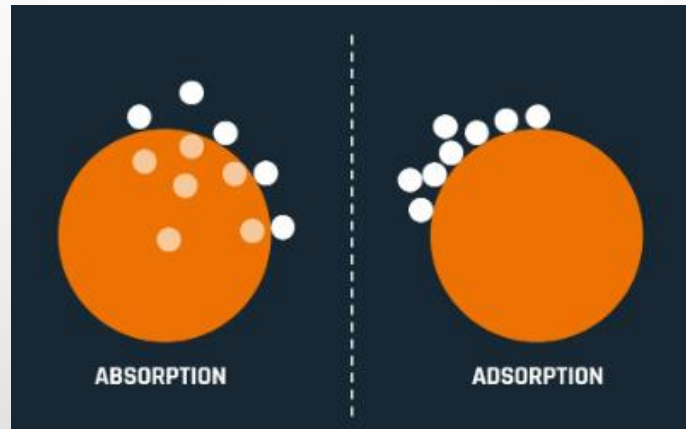
1. Solid/ gas (S/V) (Solid surface); e.g. table top exposed to air
2. Solid/ Liquid (S/L); e.g. suspension

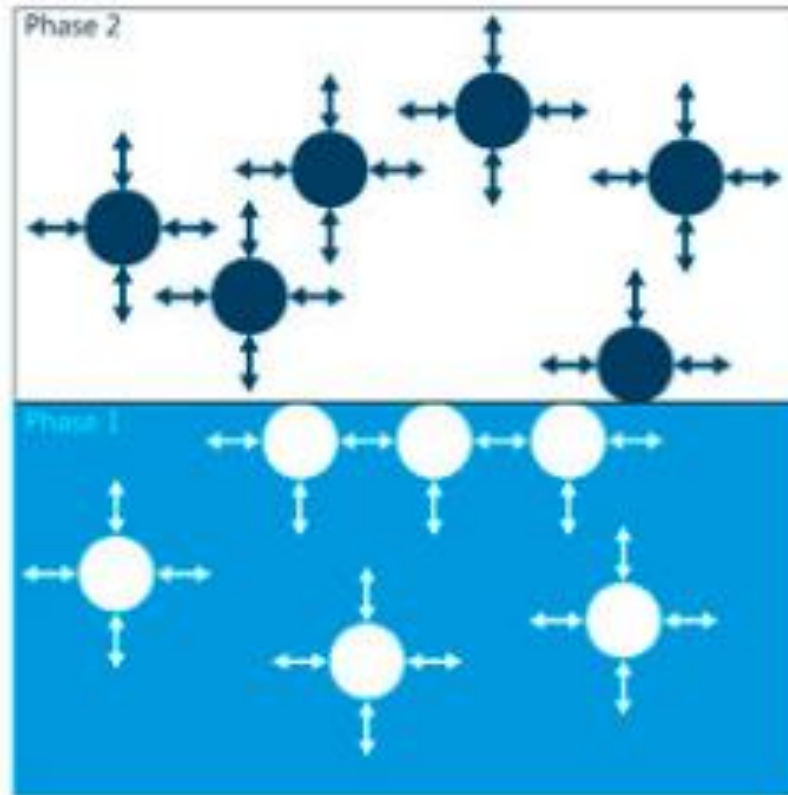
Gas/ Gas (vapor/vapor) v/v

Phase (I)	Phase (II)	Example
Gas	Gas	no interface → no interfacial tension
Gas	Liquid	Air and water
Gas	Solid	air/powder
Liquid	Solid	suspension
Liquid	Liquid	Emulsion
Solid	Solid	powders of particles

Importance of Interfacial phenomena in Pharmacy

- 1- Adsorption of drugs onto solid adjuncts in dosage forms
- 2- Penetration of molecules through biological membrane
- 3- Emulsion formation and stability
- 4- The dispersion of insoluble particles in liquid media to form suspensions





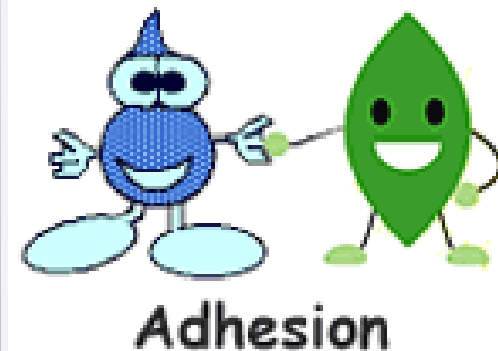
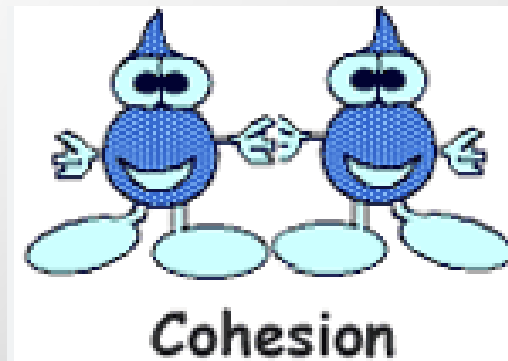
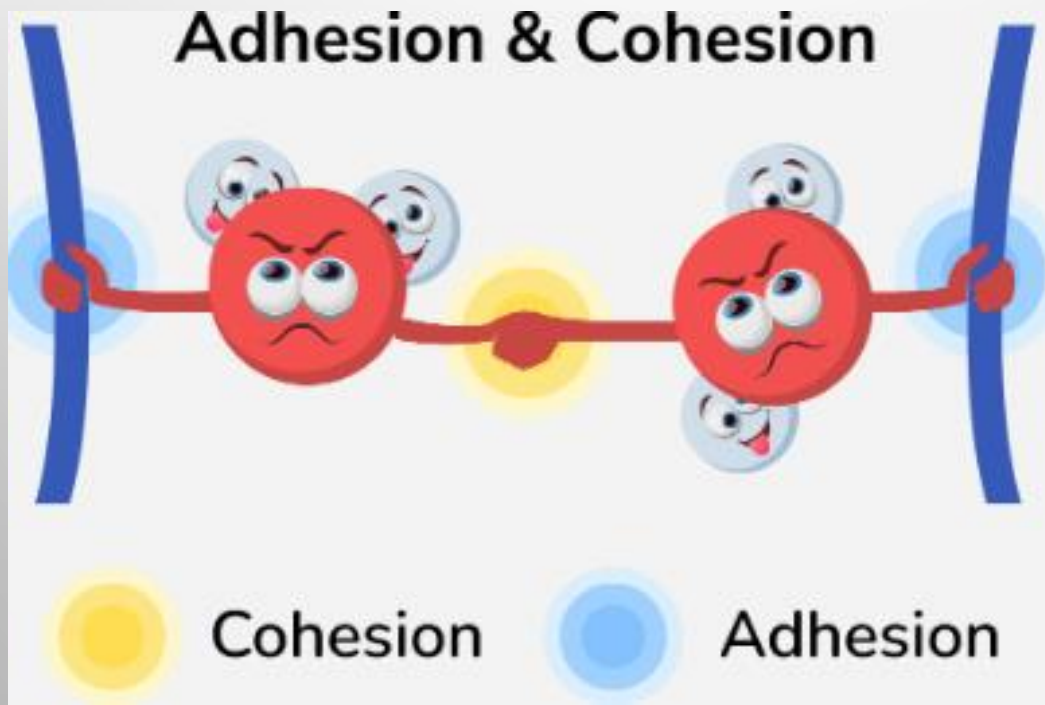
Interfacial phase:

is a term used to describe molecules forming the interface between two phases which have different properties from molecules in the bulk of each phase.

Forces of Attractions

COHESIVE FORCE: ARE THE FORCES THAT EXIST BETWEEN MOLECULES OF ONE PHASE (I.E. BETWEEN LIKE MOLECULES).

ADHESIVE FORCE: ARE THE FORCES THAT EXIST BETWEEN MOLECULES OF TWO DIFFERENT PHASES (I.E. BETWEEN UNLIKE MOLECULES).



MISCIBILITY: (Between two liquids)

Miscibility: occurs when **cohesive forces** are less than **adhesive forces**, e.g. water and alcohol.

IMMISCIBILITY: occurs when **cohesive forces** are more than **adhesive force** e.g. water and oil

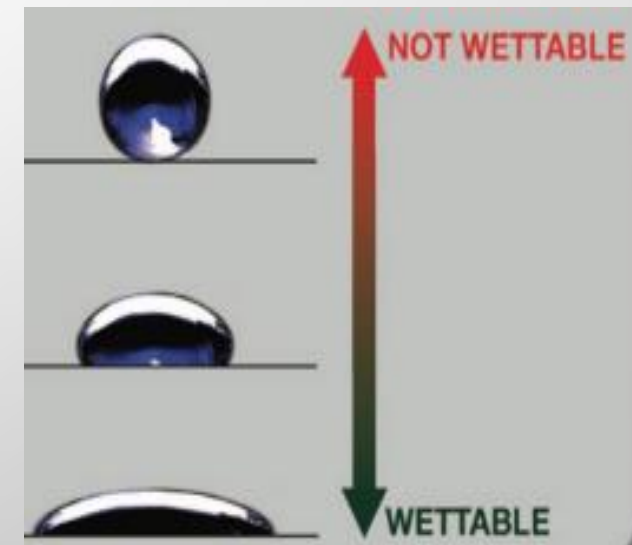
WETTABILITY: (Between liquids and solids)

when **cohesive forces** are less than **adhesive forces**

The liquid tends to wet the surface

when **cohesive forces** are more than **adhesive force**

The liquid tends to refuse the surface



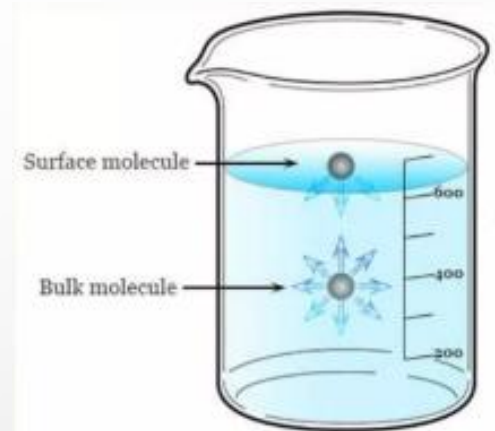
[Cohesive force > Adhesive force]	→	<u>immiscibility occur</u>	→	(water / oil)	→	emulsion
[Adhesive force > Cohesive force]	→	<u>miscibility occurs</u>	→	(water / alcohol)	→	solution
[Adhesive force > Cohesive force]	→	<u>wettability occurs</u>	→	(water / solid)		

Liquid Interfaces

Molecules in the bulk liquid are surrounded in all directions by other molecules for which they have an equal attraction (Only cohesive forces)
(Net force equal zero $F=0$) (Stable)

Molecules at the surface can only develop cohesive forces with other molecules that are below and adjacent to them; and can develop adhesive forces with molecules of the other phase
(Unstable with high energy)

This imbalance in the molecular attraction will lead to an inward force toward the bulk that pulls the molecules of the interface together and contracts the surface with a force F , resulting in a surface tension
— To keep the equilibrium, an equal force must be applied to oppose the inward tension in the surface



SURFACE TENSION AND SURFACE FREE ENERGY



SURFACE TENSION (γ)

In the surface of a liquid, there is no molecules to counteract the forces of attraction exerted by molecules in the bulk for molecules in the surface.

The inward force of attraction exerted on the surface molecules results in the tendency of the surface to contract.

This inward pull tends to diminish the surface area, and in this respect a liquid surface resembles a stretched elastic membrane

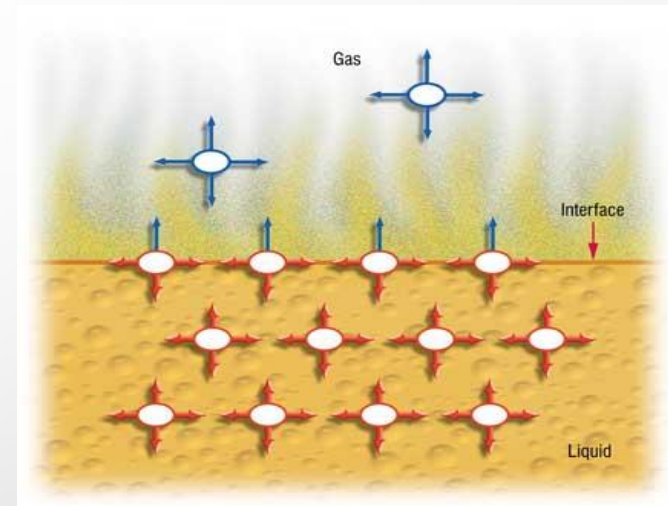
Thus the liquid squeezes itself together until it has the locally lowest surface area possible

These centrally directed forces cause the droplet to assume a spherical shape, thereby minimizing both the free energy and surface area as a sphere has the smallest possible surface area to volume ratio.

SURFACE TENSION AND SURFACE FREE ENERGY

surface tension

defined as force per unit length that must be applied parallel to the surface so as to counterbalance the net inward pull. It has the unit of dyne /cm.



Difference between Surface tension and interfacial tension

	Surface tension	Interfacial tension
Definition	→ its force applied to parallel <u>to surface</u> to counter balance (inward pull)→ <u>force/unit length</u> → its work required to create a unit of surface (surface free energy) $W = \gamma \Delta A$	its force /unit length exist at <u>interface between 2 immiscible liquid</u>
Found in	Between liquid and air	Between 2 immiscible liquid
Units	$\frac{\text{dyne}}{\text{cm}} = \frac{\text{erg}}{\text{cm}^2}$	

✓ **Interfacial tension less than surface tension, Why?**

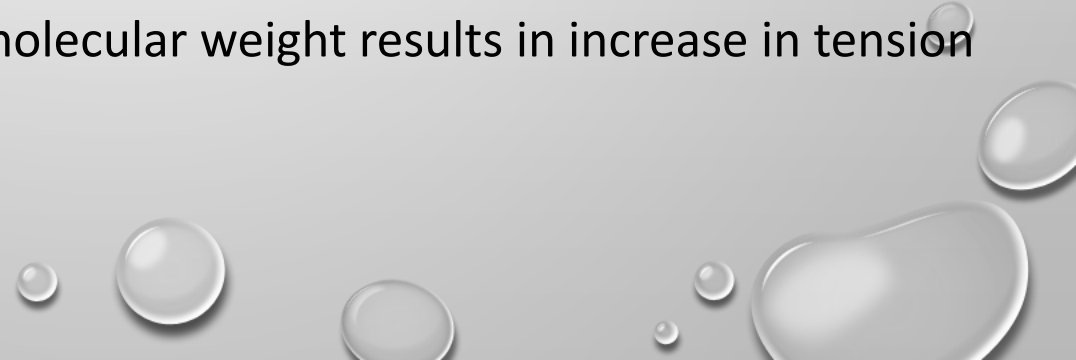
Because adhesive force between 2 immiscible liquids are greater than that between liquids and gas phase exist together

✓ **What is the tension between water and alcohol?**

No interfacial tension between water and alcohol because both are miscible and there are no interface between them

✓ **What is the relation between intermolecular attraction force and tension?**

The increase in the H- bonding or molecular weight results in increase in tension



Factors influence the surface and interfacial tension

- **Internal Factors** : The surface tension is observed in liquids due to Intermolecular attractive forces.
- **External Factors** : When substances are added to liquids, these alter the surface and interface tension. Example are Surface active agents decreases Surface tension
- **Effect of Temperature**: As temp increases the Surface tension of liquids decreases due to enhance kinetic energy that weakens cohesive force. At critical temp Surface tension is 0.

MEASUREMENT OF TENSIONS

- Methods for measuring surface and interfacial tension

- 1- Capillary Rise Method
- 2- DuNoüy Ring Method (Tensiometer)
- 3- Drop weight method (Stalagmometer) (As in Practical lab)

- The choice of the method for measuring surface and interfacial tension depend on:

- 1-Whether surface or interfacial tension is to be determined.
- 2-The accuracy desired.
- 3-The size of sample.

CAPILLARY RISE METHOD

This method cannot be used to obtain interfacial tensions.

Capillary tube is placed in a liquid sample



Liquid sample rises up tube to certain distance



Measuring the rise of liquid sample



Determine the surface tension of the liquid

• WHEN A CAPILLARY TUBE IS PLACED IN A LIQUID CONTAINED IN A BEAKER, THE LIQUID RISES UP IN THE TUBE TO A CERTAIN DISTANCE.

BY MEASURING THIS RISE IN THE CAPILLARY, IT IS POSSIBLE TO DETERMINE THE SURFACE TENSION OF THE LIQUID USING THE FORMULA:

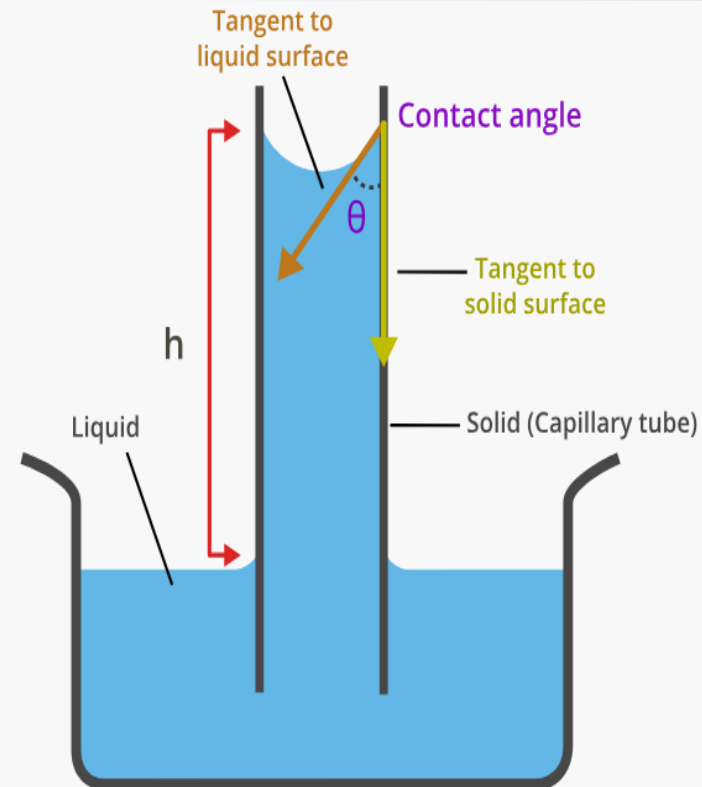
$$\gamma = \frac{1}{2} r p g h$$

r: RADIUS OF CAPILLARY

h: HEIGHT

p: DENSITY OF THE LIQUID

g: ACCELERATION OF GRAVITY



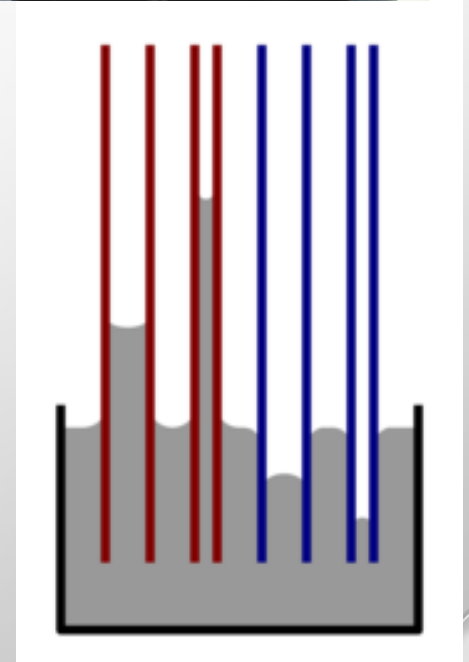
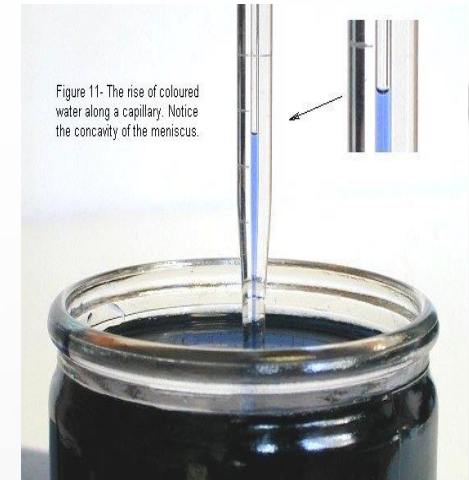
CAPILLARITY

THE RISING OR THE LOWERING OF THE LEVEL OF THE LIQUIDS INTO THIN TUBES IS NAMED CAPILLARITY.

IT IS DUE TO WETTABILITY I.E. THE ADHESIVE FORCES - BETWEEN LIQUID AND CAPILLARY TUBE WALL IS HIGHER THAN COHESIVE FORCES. SO, THE LIQUID WETS THE WALL AND RISES IN THE TUBE.

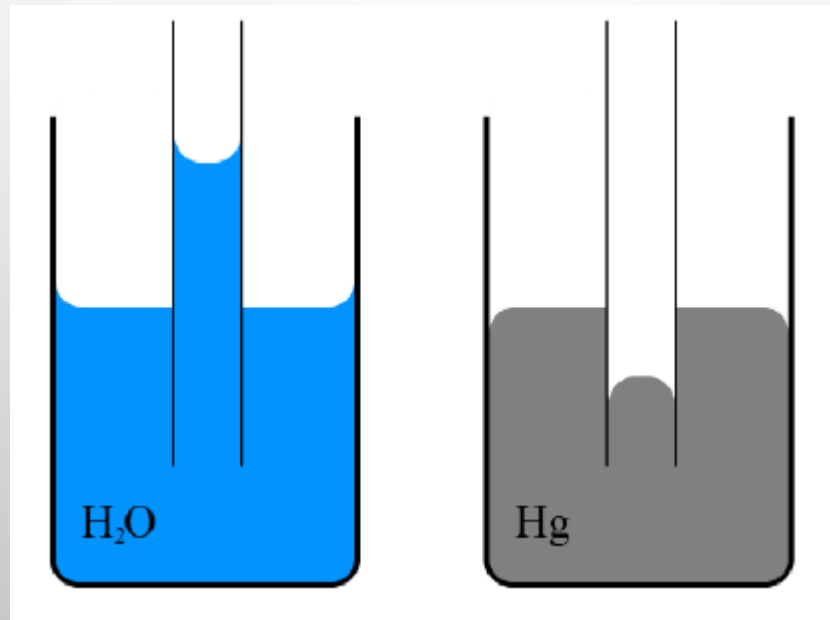
Liquid rise up until

upward force (capillarity) = downward force (weight and gravity)



- In case of water → [Adhesive force > cohesive forces] → liquid will wet capillary → liquid rise in tube (water surface tension 72.8 dyne/cm)
- In case of mercury (Hg) → [Cohesive forces > Adhesive force] → no wettability → no capillarity (mercury surface tension = 476 dyne/cm)

LIQUIDS WHICH WET THE WALLS MAKE CONCAVE SURFACES (EG: WATER/GLASS), THOSE WHICH DO NOT WET THEM, MAKE CONVEX SURFACES (EG: MERCURY/GLASS).



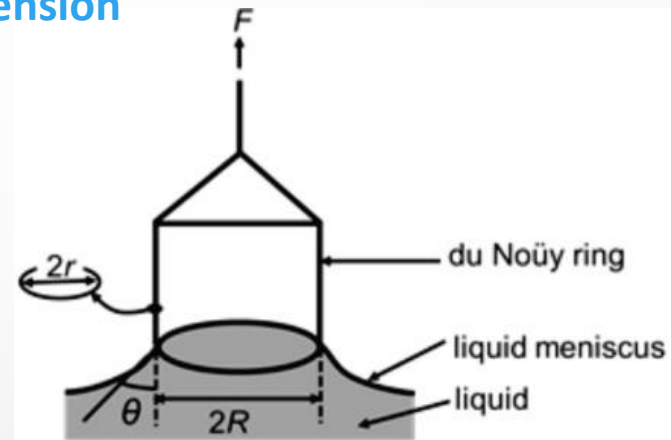
DUNOÛY RING METHOD (TENSIOMETER)

Used to measure surface tension and interfacial tension

- TENSIOMETER DEPEND ON THE PRINCIPLE THAT THE FORCE NECESSARY TO DETACH A PLATINUM–IRIDIUM RING IMMERSED AT THE SURFACE OR INTERFACE IS PROPORTIONAL TO THE SURFACE OR INTERFACIAL TENSION.
- THE FORCE OF DETACHMENT IS RECORDED IN DYNES ON A CALIBRATED DIAL THE SURFACE TENSION IS GIVEN BY:

$$\gamma = F / 2 \pi (r_1 + r_2)$$

WHERE: F = THE DETACHMENT FORCE R1 AND R2= THE INNER AND OUTER RADII OF THE RING.



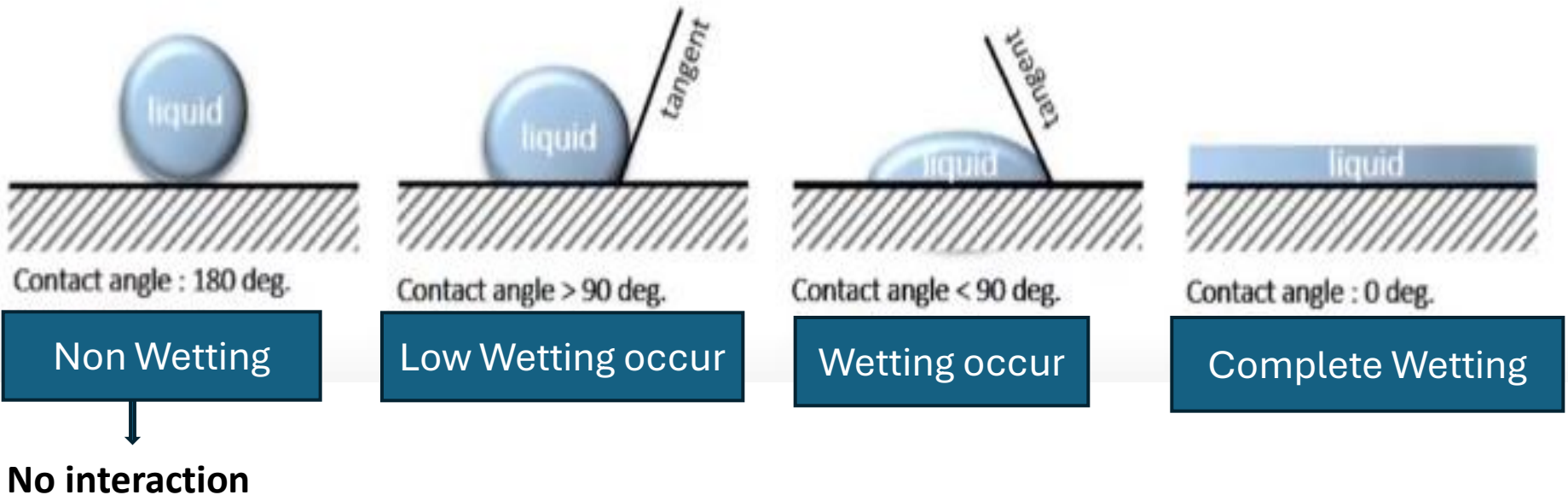
The surface tension and consequently the contact angle of the liquid are very important in

- 1- Preparation of suspension wetting of insoluble powders so as to prepare suspension
 - 2- Preparation of emulsion, mixing of immiscible liquids
 - 3-Granulation, prior to tableting involve the mixing of the powder with a liquid binder
 - 4- Film coating of tablets
 - 5- Successful dissolution of tablets or capsules necessary for penetration of liquid into pores of dosage form
 - 6- Spreading of lotion on the skin surface
 - 7- Spreading of ointment in the eye
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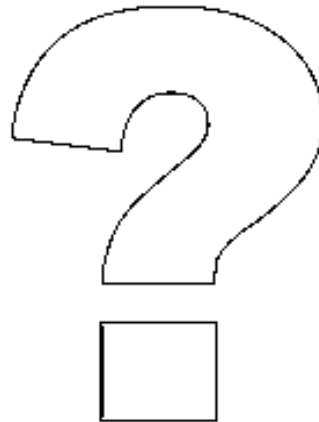
Contact angle and spreading

Angel at which liquid/air interface meets the solid

- 🔥 The physical and chemical behavior of liquids cannot be understood without taking surface tension into account
- 🔥 It governs the shape that small masses of liquid and the degree of contact liquid can make with another substance
- 🔥 to perform the previous pharmaceutical process, we need to reduce the surface tension so as to reduce the contact angle and increase the spreading of the material



**HOW CAN WE REDUCE THE
SURFACE TENSION????????????**

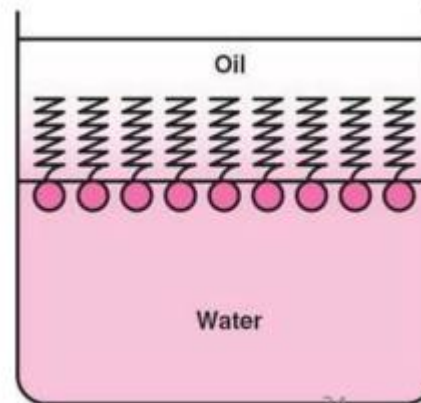
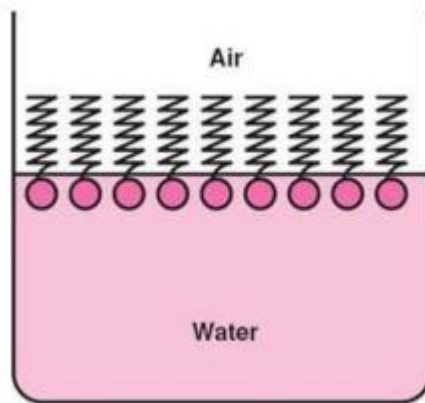
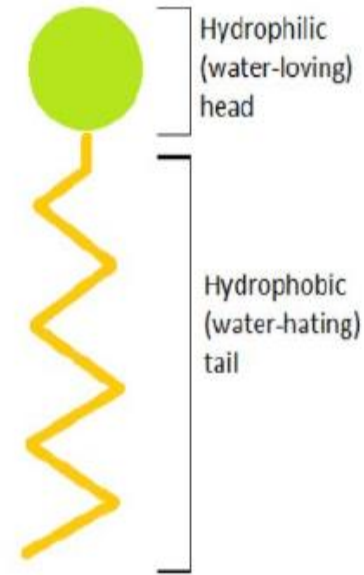


Surface active agent (SAA) (Surfactant)

Surfactants have two distinct regions in their chemical structure, one of which is water-liking (hydrophilic) and the other of which is lipid-liking (lipophilic). These molecules are referred to as amphiphile.

The balanced amphiphilic nature of S.A.A. is the character that cause them to be adsorbed at the interfaces (liquid/gas or liquid/liquid).

When such molecule is placed in an air-water or oil-water system, the polar groups are oriented toward the water, and the nonpolar groups are oriented toward the air or oil.

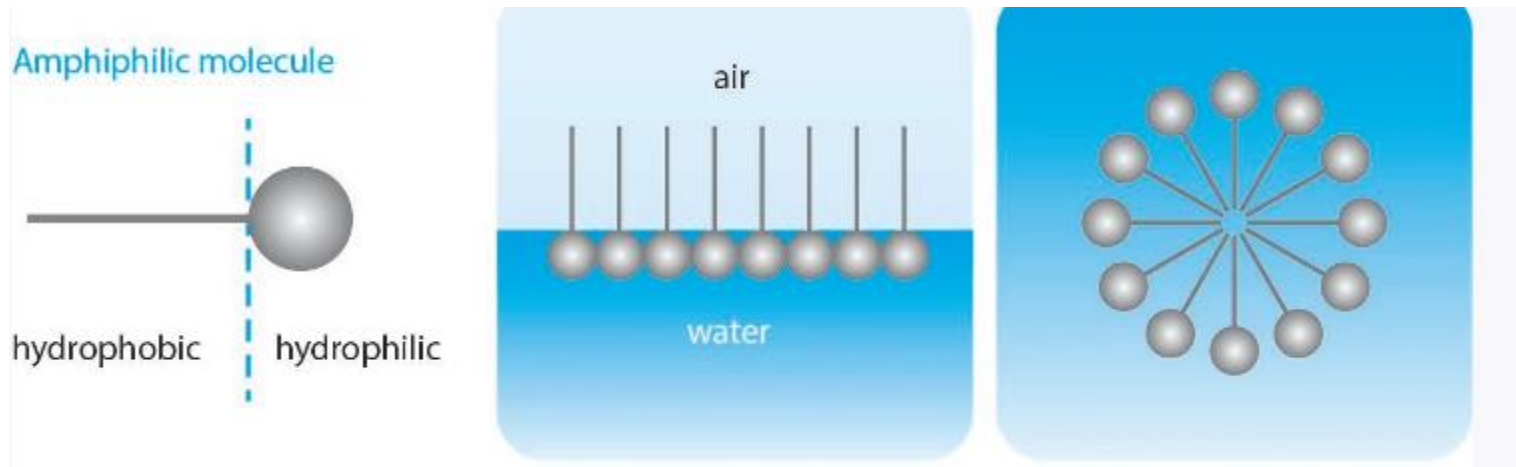


Role of a Surfactant

Surfactants have the tendency to **gather around the interface between two immiscible phases**

They are able to **reduce surface or interfacial tension** between two phases

They Serve **as good mediator** between two phases like oil and water



In the next lecture, we will explain in detail the following topics:

Surfactant classification

HLB system

Micellisation



THANK YOU!